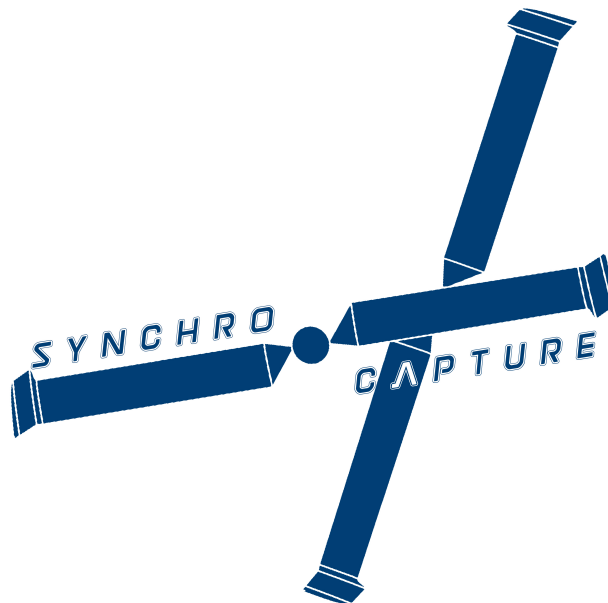


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Unmanned Helicopter for Aerial Recovery

Individual Report



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Submission Date: 15/06/2020

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Abstract

This report covers my contribution as a project manager to the conceptual and preliminary design of an unmanned synchrocopter for aerial recovery of wholly reusable launch system, designed in parallel by the WRL team and comprising of two stages.

Regarding the internal management strategy, we implemented a flexible approach to adapt quickly to new and dynamic circumstances and a low-hierarchy-level and centralised communication, then conferred relative independence and trust to individual team members. At the same time we strove for an effective collaboration with the other team, encouraging independent exchange of information with no intermediaries. Both strategies proved successful.

I calculated the base price of the helicopter as \$ 3,520,000; then I compared a traditional regression analysis on past helicopter data to neural network predictions to determine its direct and indirect operating costs as 892 \$/flight hour and 300,640 \$/year, respectively.

Finally, a qualitative noise analysis was performed to address future steps to be taken and a rough value of the Effective Perceived Noise level (EPNdB) was found as 57.7 dB, in compliance with FAR 36.

Acknowledgements

I would like to thank the supervisors, Dr Levis, Dr Montomoli, Dr Ribera Vicent, Dr Sharif Kho-daei, Dr Wynn, for their participation and feedback throughout the project. A special thank goes to Dr Buxton, for his precious advice and support to the coordination subteam.

I am also grateful to the individual members of our team: the project would not have been the same without their passion and their commitment to this complex challenge, their positive attitude and incredible skills. I hope to have been supportive to them as they have been to me.

WRR Team Members

Table 1: Team Members and Roles.

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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1 Introduction

This report outlines my personal contribution to the Group Design Project on the design of a Wholly Reusable Launch System, as coordinator of the Recovery Vehicle team. There are three main sections, team management, cost estimation and noise qualitative analysis: their length reflects their relative importance and the time devoted to each.

The management section is a narrative of the key events and decisions of the project, which were shared with the other two project coordinators. The content overlaps with theirs because every choice was made together and it would have had little sense to arbitrarily split the responsibility of specific decisions among the three individually: the reports, however, were written independently and express different takes on the strategy and decisions undertaken. It should also be considered that the project was run during the lockdown adopted in the UK for the COVID-19 pandemic, with team members often located in their home countries in different time-zones. This remarkable difficulty was not particularly emphasised in the following sections, but it should be specified that all meetings and presentations were held online.

2 Team management

The peculiarity of the project consisted in its two different scales: large for a university environment, yet small compared to industry. Because of this relatively small scale and of our acquaintance with most individual members from the three years spent together in our course, we did not rely on management textbooks for organisation purposes, which are best suited for industry: we based our decisions on common sense and engineering good practice, customised depending on the specific situation within the team.

It appeared immediately clear to the management subteam that, given the relatively high number of people for a university project, the interdependence with the WRL team and the limited time-frame, a precise plan and schedule had to be devised since the very first days of the design process. Gliner, Ghenima and I all agreed that important management and design decisions had to be made together [1, 2], and there has been no immediate distinction between our tasks: this is part of a more general approach based on common decision-making, flexibility and agility in the team structure. This allowed us to quickly respond to unexpected events and reallocate people to adapt to the current needs. This strategy only works if the coordination team is in almost perfect agreement in key decisions, which was indeed the case. We found common grounds both in the attitude to keep throughout the project and in the goals of the team as a whole, which allowed us to focus on fluid, practical action rather than time-consuming internal discussions. The approach can be described in terms of the project objectives, its practical limitations and the resulting plan of action.

2.1 Goals and limitations of the project

The main goal was to have a functional design at the end of the five weeks, while trying to maximise the performance (and consequently the mark) of the individual members. The philosophy behind our design approach consisted of having an innovative and ambitious design that would consider, however, the feasibility and likelihood of success of the overall mission. We wanted the project to be interesting under an academic point of view and, at the same time, be feasible if it were to be actually implemented in real life. Our goals had to be achieved keeping in mind a series of limitations: first, five weeks gave us a very tight deadline to deliver a good design, which made us accept suboptimal choices throughout the project to make sure we could still deliver reasonable results. We realised that in a first iteration we could not optimise certain parts of the vehicle without either sacrificing later important calculations for lack of time or keeping an overwhelming schedule for all team members, so we opted for making

as many decisions as possible as we progressed, excluding right away rewarding but possibly riskier alternative configurations. An example is given by disregarding a tiltable back-rotor for the helicopter, as explained in detail in Section 2.4. Moreover, in order to allow people to work on their report, we specifically aimed at finalising the design earlier than the five weeks provided so that a few days could be devoted to report writing. The number of days had to be flexible to account for the different timelines of the single subteams, this discussion will be expanded later and is an example of the agility we strove for.

In fact, a dynamic approach allows quick changes in strategy depending on individual needs and feedback, which in our case was given by the lecturers either in general meetings or to the individual sub-teams they supervised: the aim was to eliminate or reduce the impact of any imperfection as soon as it was spotted. Any negative feedback or aspect of the design was addressed as soon as possible, whereas we focused on keeping and further improving all features that drew positive comments.

2.2 Internal cooperation and trust

Along with agility, two other aspects we focused on were cooperation among team members and trust. The former was deemed essential for a good outcome of the project as a whole: we tried to set up a network of communication early in the design phase, so that only few corrections were needed in the following weeks. The launcher team set up a series of communication channels on Microsoft Teams, Slack and Trello, as well as a shared database on Valispace (a platform where key parameters of the design are stored and updated): for the scope and extent of the project, we considered this as inexplicably overcomplicated, as people would not have had the time to get accustomed to all the platforms and we would have risked losing key information. Centralising the workspace was the best and least dispersive solution, and Teams was the obvious choice of channel to run the project effectively. Moreover, we had to ensure key data was communicated immediately to the relevant subteams: this was tackled first by nominating subteam leaders, who were supposed to act as intermediaries between teams, but after the first two weeks we naturally pushed for an independent communication among team members by assigning specific roles and tasks within working groups. These were created as a way to manage the design of macrofeatures, like the fuselage or the rotor, or broad topics, such as performance or the alternatives for vehicle choice in the first week, and were formed by members from different subteams as an example of inter-team cooperation. They proved quite effective and received excellent feedback as a way to split the work: in particular, Dr Buxton praised the structure of the presentations, that were organised in a similar manner. Having effective communication at low-hierarchy levels allowed us to avoid whole team meetings as a form of design updates: the launcher team adopted them, initially quite frequently, then less and less as the pace increased. In fact, although perhaps more direct and easier to manage, they take away useful time from everyone's schedule, which could be used otherwise for calculations and practical design improvements. Throughout the different design phases, we faced and solved communication problems with the final goal of having a perfectly functional network within our team and that everyone had a global vision of the helicopter characteristics, so that they would have been able to independently spot mistakes and avoid delays due to incorrect data. We particularly stressed the importance of informing the team leaders of any results or changes: even when allowing team members to be fairly independent, we had to be notified of any relevant updates in order to adapt the team's strategy to the new situation and plan in advance the following steps. Problems related to this matter are detailed in Section 2.4. All things considered, although not perfect, such an organisation allowed a quick flow of information with minimal input from the coordinators, which ultimately is part of the following aspect of our strategy.

In fact, we aimed to build an environment of trust between us and the team members: this had to occur spontaneously, determined by our attitude towards them and their work. A relaxed and cooper-

ative environment is notoriously more productive than an oppressive atmosphere, where decisions are imposed rather than shared with the team as a whole. Therefore, we immediately decided to give freedom to each subteam to make their own decisions and only interceding had a conflict arisen, whether among themselves or with other groups. In this way we recognised the relevance of their work and the fact that they were more informed than the three of us on the specific decision to be made. This approach eased the workload on us and made the individual members more independent and proactive, without the frustration of being dictated what to do step by step. Obviously, we stepped in to make some general decisions where a global trade-off was involved and speed up the process in some cases where the debate was unproductive or results were quickly needed: however, every decision was made after all options were evaluated, keeping the subteams involved at all times. Also, we accepted the risk that such freedom could lead to potentially incorrect results from some people, if not supervised: to address this issue, we encouraged cross-checking within subteams and working groups and inspected more closely the work being done when necessary.

The next subsections will explore in detail the vehicle choice and the following project timeline, including key engineering decisions and specific problems we faced along the way, and finally on how we cooperated with the launcher team.

2.3 Vehicle selection

As mentioned previously, the first week was dedicated to the vehicle selection and the related recovery method. Some very unconventional designs were proposed, such as a large number of drones at the edges of a net, an electromagnetic decelerator, a cannon net or even flying wings that would attach to the descending stage and make it land: none of these were clearly feasible. It soon became clear that the choice had to be made amongst three categories, an aircraft, a VTOL or a helicopter, and the first week was devoted to a trade-off study, waiting for a first approximate design of the launcher to confirm the feasibility of our choice. As coordinators, each of us took the lead of one of the studies: we split the team into three main working groups, each with members from different subteams to have a global understanding of how the vehicles would have been at a later design phase. The division was voluntary, but we made sure that roughly an equal number of people from each subteam would join different working groups (so that the Aerodynamics subteam was not entirely in the VTOL group, say). This facilitated a balanced selection of people presenting in the following meetings.

Personally, I took the lead of the VTOL group. Alongside my general organisation duties, I started to implement the strategy mentioned earlier, intervening only when necessary and letting the group investigate advantages and disadvantages of our alternative. This allowed me to focus on the communication with the launcher team and on planning the next steps with the other coordinators. I partially facilitated the sharing of resources between the VTOL and the helicopter concept groups, given the similarities between some design features, and devised a correction factor for the Research and Development (R&D) costs of both, depending on their TRL: the method is explained in Section 3.1 and was used to crudely compare the two vehicle types.

Regarding the final choice, the fundamental metric adopted was the accuracy of the landing location provided by the launcher team: given a twenty-minutes recovery window (from the moment we knew the exact landing point with respect to the boat to the beginning of our manoeuvre), the helicopter was best for distances up to 50 km, whereas the aircraft for those greater than 150 km. The latter had, however, a series of problems, including high costs (calculated by Sharma [3]), little manoeuvrability and few attempts available, which made the VTOL a more appropriate choice for distances between 50 and 120/150 km. The difference in TRL between helicopter and VTOL resulted in a difference in cost, but this wasn't large enough to dictate a firm choice on its own. The launcher team confirmed that they could indeed achieve a precision of 50 km, leading to the helicopter being chosen. The proper

trade-off was finalised by means of an Analytical Hierarchy Process (AHP) matrix, frequently used in the reports of the University of Maryland submitted for the Vertical Flight Society (VFS) Student Design Competition [4], and a risk matrix: these are included in Appendix A.2, along with the key results obtained.

2.4 Project timeline, key decisions and relevant issues

After the first week, the focus shifted towards the baseline configuration and the initial sizing of the helicopter: we opted for an unmanned synchrocopter in the second week. The main design decision here was not to include a tiltable back-rotor, able to act as additional lifting and pitching device when horizontal (which would have greatly helped counteract the pitching moments arising during the recovery manoeuvre) and as a pusher when tilted, giving us an edge when a fast travel to the recovery location was needed. It was optimal for most teams except for Flight Mechanics, that would have encountered serious difficulties in modelling it. We agreed that a suboptimal design that could be modelled by the end of the project was better than a well-performing vehicle that could not, so we went for a H-tail that would have given enough pitch control anyway.

The unconventional nature of the helicopter led to delays in its initial sizing. Under this perspective, we could not follow strictly the suggested schedule of two weeks to finalise the conceptual design: firstly, one week had already been devoted to vehicle selection, that had subtracted resources away from the helicopter concept; secondly, the conceptual and preliminary design phases were intrinsically tied together, given that the initial sizing required an evaluation component-by-component [5, 6]. Therefore, we formed parallel working groups, following slightly different timelines but able to rapidly change input values as they were being updated by other groups. Towards half of the third week, we finalised the weight estimation after an iteration involving fuselage, rotor, engine, tail and obviously weight groups. We began to face issues related to the aerodynamic rotor design, as the two members involved struggled to communicate effectively within the team. I requested them to inform more frequently other people of their results, with positive answer but no visible effect. We received values of key features days after they were calculated, and only after requesting the values specifically.

From the fourth week onwards we were in a well-defined preliminary design phase, with numerous groups cooperating together very well: we had to guide the choice of the optimal fuselage design, as at first the drag contribution of the inlet was not being considered. This was a minor issue, resolved promptly, whereas a more urgent problem was found as a mistake in the calculations of the rotor diameter: a disk loading of 6 lbs/ft^2 was initially arbitrarily chosen, but the disk area was determined incorrectly, leading to a shorter radius than required [7]. The two options were to either repeat the whole sizing iteration procedure or recalculate the new disk loading and verify this was still within an acceptable range for a helicopter. With a new value of 9.4 lbs/ft^2 , we accepted this latter alternative to not disrupt everyone else's work, but well aware that the efficiency of the design would have been heavily compromised. Towards the end of the fourth week, we encountered new issues with the blade design: the aerodynamic optimisation approach was being delayed and we were not clearly informed of how it would work, due to lack of communication. Instead of sending another message, we had a face-to-face meeting, which seemed to have the desired effect of making them decide sensibly key parameters rather than letting a long computational process do the work without proper trade-offs involved. Their design was finalised the day after, but we suspect the blade still has issues to be resolved, as it has been indicated by the Flight Mechanics and Simulations teams as a possible source of inefficiencies.

In conclusion, we managed to finalise the main design choices for most subteams on the Monday of the final week, leaving up to Thursday for minor improvements: this would have allowed the Simulations and Flight Mechanics subteams to have certain values for their analyses. Furthermore, a working

group on Performance was created and we expected the design to be final on the last Saturday: to help members from these subteams, we strongly encouraged them to start writing the report earlier that week to allow some time for further work in the last few days.

2.5 Cooperation with WRL

One of the premises of the project was the intrinsic relationship existing between the two teams, for obvious design reasons: this collaboration, however, was indeed harder than expected. Initially, I took the responsibility of the communication with the other team: I joined their channel on Slack and their database on Valispace to keep track of their design parameters. The first data we needed from them was, as already mentioned, the precision during descent in order to select the vehicle. We received a positive response affirming that they were able to guarantee a range within the 50 km limit for the helicopter, but this was followed by a reply from one coordinator, showing uncertainty on the previous statement. The situation was solved only after the first presentation, where the other team confirmed this level of precision, after days when we kept asking without receiving a final answer: they referred to the previous conversation claiming they already made that clear, despite the vague reply.

To our understanding, rocket design is different in terms of the small margin of error in initial phases of the design, leading to multiple configurations being considered altogether: this is perhaps the main reason of a common issue we had, which was some constant vagueness and delay in specifying some key values such as masses, landing precision and dimensions. No values were final for the first three weeks, which made our sizing difficult and caused some days of delays in certain cases. We also found ourselves in a situation where we had to chase the coordinators to obtain values, whereas we would have expected a spontaneous exchange of information. Given the difficulties encountered in the first week and the contradictory responses on the channel, we decided to bypass the intermediaries and include some members of our team on Slack, as well: from then on, they would have got in touch directly with people from the other team, directly involved in the relevant topics and able to give quick answers. The shift to low-hierarchy-level communication, once again, proved effective and we kept such an approach for the following weeks.

Other issues arose during our initial sizing: the mass of their first stage (our constraint in terms of maximum payload) was varying between 1 and 1.4 tonnes: waiting more days would have jeopardised our schedule, so we immediately called for a meeting and, after having clarified the reasons of such a high uncertainty, we opted for the safest constraint, at the potential cost of over-engineering the helicopter. We also were not informed directly of some relevant decisions on their end: we discovered that the fairings would have stayed attached to the second stage only indirectly through their chat, and in the very last week we weren't notified of changes in the mass and length of the upper stage, which resulted in some of our team members using outdated values, without dramatic consequences. For further and more specific details on interactions between the two teams, Gliner [1] described the orbital pass concept, devised for a better business case, and Jenner [8] dealt with the recovery manoeuvre and therefore the interface with the parafoils used.

On our end, some mistakes were made especially in the first week, being perhaps impatient in asking for values and not having a global idea of how their team was organised and how their design process was being carried out. This led to unnecessary irritation in some cases, which is not beneficial for a cooperative and productive environment.

We are aware that the launcher project was clearly more independent from us than we were from them, but we had the feeling that the recovery vehicle was considered as secondary, even in the first weeks where the coordination between the teams should have been greater. In retrospect, a better organisation of the projects as a whole would have probably been to designate a unique team of six coordinators for the two projects, so that it would have felt more like a unique design process.

Moreover, a remarkable issue for our team was to be in a more fast-paced timeline than theirs: under this point of view, it would have made more sense to design the recovery vehicle after an initial sizing of the rocket, so that the two timelines approximately coincided. However, despite the abovementioned misunderstandings, most of the interactions with WRL were fruitful, also considering they decreased remarkably after the conceptual design phase was concluded.

3 Cost estimation

A major design driver for the overall mission is the cost saving with respect to letting the stages precipitate and be lost, or to devising a propulsive autonomous recovery for both stages (leading to large development costs). One goal is therefore to keep the total mission costs for rocket and helicopter below those of a common launch, aspect that will be dealt with more accuracy by Gliner [1] in the business case analysis, in conjunction with the WRL team.

3.1 Base price for VTOL and helicopter concepts

The cost estimation model proposed by Harris and Scully [9] was used to compute the base price of the helicopter and VTOL concepts in the first week, which was a metric for the vehicle selection, as explained in Section 2.3. Relevant equations and input parameters are included in Appendix A.3. The model relies on two main assumptions: first, the regression analysis performed by the authors elaborates data from ordinary, mainly manned vehicles, which little have to do with a completely novel and unconventional design; second, the base price is calculated for mass-produced helicopters and therefore would remarkably underestimate the price of a vehicle operating in a market niche. The business case will consequently also consider the possibility of reconversion for other missions [1]. Moreover, Harris and Scully's predictions are accurate within 20% of the actual value, for 96% of the helicopters considered. Despite the model being of remarkable accuracy, in our case these factors introduce uncertainty, whose effect cannot be easily determined in a preliminary design phase and will impact possibly more the estimation of operating costs in the next section.

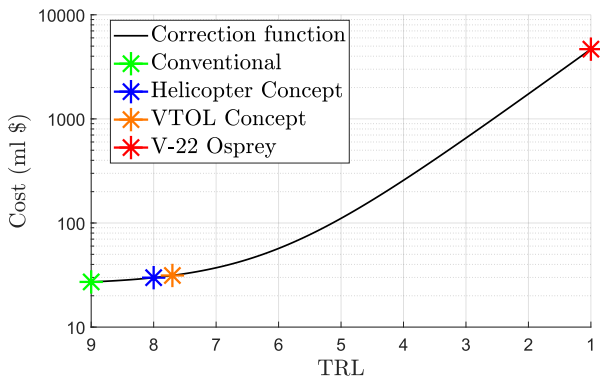


Figure 1: Corrected R&D cost by means of an exponential interpolation in function of the overall TRL.

behaviour. As shown in Figure 1, we interpolated a function of the form $y = Ax + B$ so that a TRL of 1 corresponded to the V-22 Osprey, having a disproportionately and notoriously high R&D cost of 12.7 bn \$, whereas we considered the price calculated using Harris and Scully for a TRL of 9. The resulting corrected R&D costs for VTOL and helicopter were estimated as 36 and 69 ml \$ respectively. As expected, the helicopter had a large edge on the VTOL concept, but still not sufficient to exclude the latter. It should be noted that this analysis is not meant to be accurate in any way: there are

However, a VTOL is clearly less conventional than a helicopter, which translates into generally higher development costs, that for the purposes of this analysis were assumed equal to roughly four times the base price, from the reports of the University of Maryland [4]. I introduced a correction factor dependent on the TRL of each configuration, which was estimated as 8 and 7.6 for helicopter and VTOL, respectively, taking an average of the TRLs of relevant individual sub-systems. Alexander [10] suggests that costs increase exponentially with System Hierarchy (SH) level: this is not equivalent to the TRL, but we evaluated the latter according to a similar grading system, which allowed us to assume a similar

several inexcusable assumptions that cannot be implemented in a more advanced design phase, but we accomplished the goal to quantify a comparison of costs for two very general concepts, which was eventually considered when selecting the final one.

3.2 Base price of the final configuration

Harris and Scully's model was adopted again for the final design and returned a base price of \$ 3,520,000, which was in line with typical values of helicopters with similar MTOW. The correction for TRL was not implemented, as the configuration was deemed in line with current technology levels. As a matter of fact, we are confident that a similar price can be achieved given our vehicle, by paying particular attention to material and configuration trade-offs in some subsystems to minimise their cost. Team members were often reminded of cost considerations throughout their work and this led to major trade-offs in specific aspects of the design: the main ones are found in the fuselage truss and the engine. The former, as explained in more detail by Sharma [3], was either going to be built out of aluminium or CFRP: the latter was much more expensive than aluminium, both as material *per se* and for manufacturing the shapes and most importantly joints of the structure. This was a main driving factor in the selection of aluminium for the truss structure, despite a penalisation in terms of weight. The engine selection, instead, mainly focused on the choice between conventional and electric engines, on the cost difference between a single or double spool turboshaft: as detailed by Arangath [11], the latter was chosen despite a slightly higher production cost, due to its higher complexity than a single spool, thanks to a lower fuel consumption which in turn reduces considerably the operating costs of the helicopter, fundamental for minimising mission costs. These have been supported by traditional choices in other areas of the design (e.g. material selection for the blades, the transmission layout etc), which are common to most helicopters and therefore will not have a negative impact on the price compared to average.

3.3 Operating costs

The operating costs of a helicopter are usually defined as direct or variable and indirect or fixed: direct operating costs (DOC) are proportional to the number of flight hours of the vehicle and include expenses such as for fuel and maintenance, whereas indirect operating costs (IOC) are fixed and include costs like insurance, depreciation, servicing and crew salaries [12]. The DOC is expressed either per year (specifying the number of flight hours) or per hour, while the IOC is always indicated on an annual basis. Estimating these parameters is not straightforward, as there is no formula equivalent to the Harris and Scully model for operating costs, and such evaluations are usually performed by specialised companies, with the time, expertise and resources to obtain accurate results. One of these, Conklin & de Decker, was repeatedly mentioned in the reports of the University of Maryland: alongside consulting for owners, it sells licenses to a software and a large database of helicopters [12], but such an expenditure was deemed excessive for the purposes of the project. Another website, Business Jet Traveler, was proven to be helpful as it gathered data for 126 helicopter models exactly from Conklin & de Decker's archives [13]. The information available was not as exhaustive as hoped for, but comprised the following relevant features: base price and operating costs, maximum takeoff weight, performance parameters such as range, maximum altitude and forward speed achievable, internal volume and dimensions, as well as number of passengers and years of start and end of production. This variety of samples was also categorised by manufacturer, for a total of twelve companies, among which Airbus Helicopters, Bell, Leonardo and Sikorsky had the largest number of models. Kaman is also worth-mentioning, as the K-Max is particularly representative of our design. Unfortunately key information (such as the number of rotors, of blades, or the installed power of each rotorcraft) was not included: since Harris and Scully's model made use of such data, they might have had some relevance

in the following analysis, but it was not possible to readily retrieve these characteristics elsewhere. Web scraping was the most immediate solution to collect the relevant data from all helicopters, and the results were automatically saved in an Excel spreadsheet. At this point, we aimed to predict the operational costs both by a conventional regression analysis and by deploying a series of neural networks. Both approaches are detailed in the coming sections, with a discussion on the drawbacks and advantages of each and on possible improvements to be made.

3.4 Choice of input parameters

From the number of parameters available as inputs, we attempted to keep only the most significant variables, so that we could reduce the complexity of the model and obtaining reasonable results at the same time. Principal component analysis was briefly considered as a way to reduce the number of inputs, but was discarded as the small number of parameters allowed a more thoughtful choice with proper tradeoffs and exclusions. The primary goal was to include only variables that were general enough to describe an unmanned synchrocopter based on data from conventional rotorcrafts: hence, number of passengers, size and internal volume were readily discarded due to our uncommon layout. Years of start and end of production were initially considered as a rough estimate of the technology level of each helicopter (the former especially) and of the quality/popularity of the vehicle (a long production period indicates it was a good product overall). However, these measures were not considered an accurate representation for all helicopters and their impact on the regression was evaluated by trial and error: the change in R^2 after removing one of these from the input set never exceeded 1%, so they were ultimately discarded.

In order to validate our resort to regression analysis, it's useful to visualise any possible dependence of the operating costs to the input variables. Plots describing their relationship with the most relevant predictors are included in Appendix A.3. In summary, the DOC is approximately linearly proportional to the MTOW and the base price, whereas there is a quasi-logarithmic relationship between the IOC and the same quantities, with the difference that this displays three main clusters rather than a continuous cloud of datapoints. Relationships with the other parameters were not so well-defined, but their inclusion led to an improvement in the model accuracies.

3.5 Regression analysis

The Regression Learner toolbox in MATLAB was used to identify the best regression algorithm for our dataset. All the models available were trained and compared with each other, aiming at minimising the root mean squared error and therefore maximising R^2 : the process was repeated for both the unmodified dataset and the same variables normalised to have zero mean and unit standard deviation. This procedure is common in regression analysis and machine learning, as it prevents a predictor with a large variance to have a larger impact on the model than others with a smaller variance: in this way the loss function is not biased towards such predictors. We cross-validated the regression model using a batch size of seven instances in order to prevent overfitting and we therefore extracted four models in total, two for each of direct and indirect operating costs, for normalised and unscaled inputs.

We can infer from these results that, in general, the DOC can be predicted more accurately than the IOC, with a higher R^2 , and is well described by a linear model in both cases. The normalised inputs are processed well by a step-wise linear regression when finding the direct operating costs: this is equivalent to an ANOVA regression, where input variables are included or excluded according to their influence on the overall variance of the model [14]. Despite the higher accuracy, both inputs and the output have to be scaled back to non-normalised quantities and therefore we can't obtain a simple formula for direct prediction. A linear regression [15] was instead the optimal choice for unscaled inputs and output, returning quite simply Equation 1.

$$\begin{aligned} \text{DOC (\$/year)} = & 71,480 + 142 \text{ MTOW (kg)} - 0.02047 \text{ Base Price (\$)} - \\ & -144.32 \text{ Range (km)} + 0.98832 \text{ Altitude (m)} + 97.666 \text{ Speed (km/hr)} \end{aligned} \quad (1)$$

The IOC, on the other hand, was determined via an exponential Gaussian Process Regression, which is a probability-based regression model [16]. This was the optimal choice both with normalised and non-normalised inputs, with a slightly lower accuracy than for DOC, which was expected, given the more unusual dependence of the indirect operating costs to the other variables, as previously shown. Due to the probabilistic nature of the model, we could not extract a straightforward equation for quick use. Hence the values of DOC and IOC were selected as 401,400 and 300,640 \$/year, respectively. Since the direct operating costs were indicated for a reference of 450 yearly flight hours (FH), it's convenient to express the DOC per hour, which would be 892 \$/FH. This is an encouraging result, especially if compared to the Kaman K-Max, which has an approximately double DOC (1846 \$/FH) and a slightly higher IOC (324,640 \$/year).

3.6 Neural network predictions

This approach consists of predicting the operating costs of the helicopter via neural nets, with the same five input variables as before and the two operating costs as outputs: this approach is more unconventional than regression analysis and offers potential benefits in terms of accuracy and modelling hidden trends in the training dataset that are ignored by a particular regression model and might be important in prediction from new inputs. A major drawback of neural networks is that they work in a somewhat obscure way and no simple formula can be determined as readily as from a typical regression model. A general structure of a neural network is designed to resemble a human brain: it's composed by one or more hidden layers, each comprising of a number of neurons which are essentially functions with variable weights. The network gets trained on a subset of the data available and the weights are optimised by an algorithm, chosen depending on the needs. This will be specified later for our case. Further complications arise because of the small dataset available to us, which increases the chances of overfitting and is heavily dependent on the selection of the training set. Overfitting occurs when the model converges to a very small error and performs well with the training set, but is unable to generalise the predictions for other inputs, resulting in a large error in these cases. Moreover, the training set is usually chosen randomly, which can make the model deviate from the correct trend if a large number of outliers is selected by chance: for example, if all the outliers of the dataset are included in the training batch they will have a larger impact than, say, half of them being selected. Our case is inevitably affected by both phenomena, which can be counteracted in the following way: the algorithm chosen in MATLAB was Bayesian regularisation, which was devised purposely to reduce the chances of overfitting on small datasets [17]; multiple training sessions were performed on the same neural network in order to obtain a distribution of predictions for our DOC and IOC, from which we could extract sensible values. Five hundred sessions were deemed appropriate for our scope, as they were sufficient to fit a representative probability distribution and were not too computationally expensive. Another particular issue is the number of layers and neurons to be chosen: there is no mathematical method but rather rules of thumb for a sensible selection, which include opting for a simple structure to prevent overfitting, especially for a small number of samples. Therefore, we trained networks with one and two layers, each with from two to six neurons. Each of these gave promising results, as shown in Figures 2 and 3, where a histogram groups the number of instances for R^2 and the predicted value of both DOC and IOC for the case involving two layers, with four neurons each. We fitted a Kernel distribution to the plot and we extracted its mean, median and standard deviation. In all cases, the median was found to be almost coincident with the mean, supporting the assumption that the data is distributed almost normally, which can also be inferred from inspection. This is not necessarily the

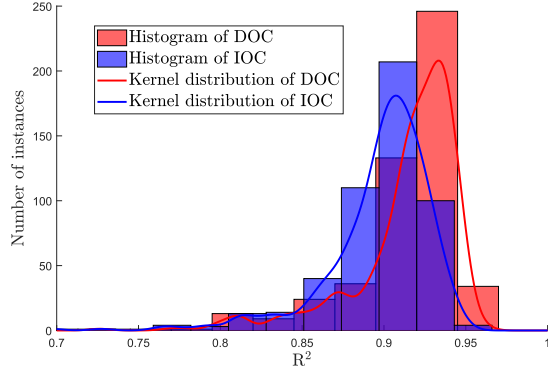


Figure 2: Histogram and Kernel distributions of the R^2 of direct and indirect operating costs for a neural network with two layers, four neurons each.

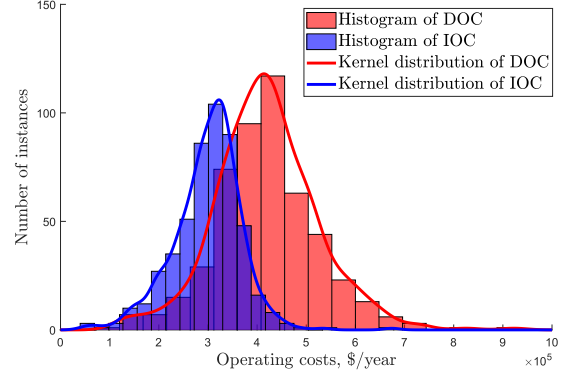


Figure 3: Histogram and Kernel distributions of predictions of direct and indirect operating costs for a neural network with two layers, four neurons each.

case, so we did not implemented such a distribution. To evaluate which network structure was best, we plotted each R^2 and related prediction for the entire dataset against the number of neurons in each layer and the best combination of high accuracy and low deviation from the mean was chosen. Networks with only one layer were discarded for their lower accuracy. Figures 6 and 5 show the plots of R^2 and predictions of the DOC for two-layer structures, highlighting the means and the values at plus or minus a standard deviation. Similar plots were produced for the IOC. A network with four neurons per layer was chosen, given its high accuracy ($R^2 = 0.92$ and 0.89) and small standard deviations for both operating costs: its related distributions were shown previously. This implies that most of its training sessions predict consistently well the trend of the entire batch (test set included) and we compared these predictions with the regression analysis results, as shown in Table 2.

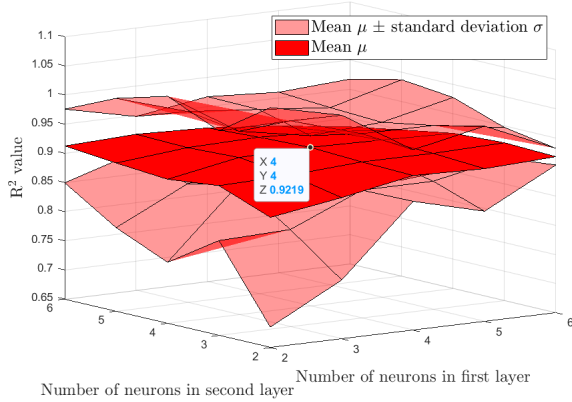


Figure 4: R^2 values of DOC predictions for two-layer networks as function of the number of neurons: the standard deviation tends to decrease as the number of neurons increases, while the R^2 reaches its highest values at (4,4) and (4,3).

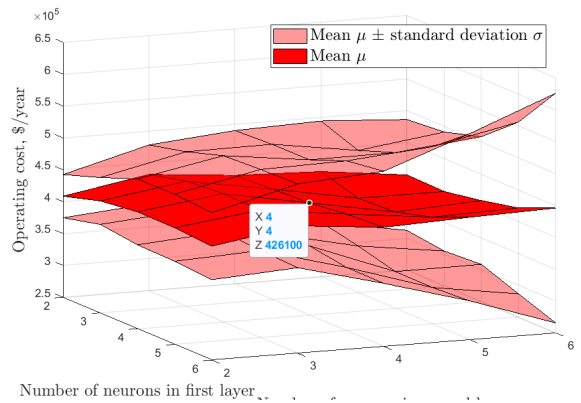


Figure 5: DOC predictions for two-layer networks as function of the number of neurons: the standard deviation increases as the number of neurons increases, with a sharp rise when they are more than five in both layers.

Although the results were quite similar, regression predictions were chosen as a final value for the operating costs, since more conventional. However, the method can be further improved, since maximising R^2 is not necessarily the optimal strategy to find the best prediction: since the RMSE is an absolute measure, the relative percentage error can still be quite high in case of small absolute values. A possible solution consists in minimising such relative error, requiring however a more advanced analysis.

Table 2: Comparison of the results from the regression analysis, the neural network study and the real costs of the Kaman K-Max. Values in bold were chosen as representative of our design.

	DOC (\$/FH)	R^2 (DOC model)	IOC (\$/year)	R^2 (IOC model)
Regression analysis	892	0.92	300,640	0.89
Neural network	944	0.92	315,100	0.90
Kaman K-Max	1846	-	324,640	-

4 Noise analysis

Noise is a fundamental challenge to be faced in helicopter design, extremely difficult to model accurately and strictly monitored according to regulations. The set of rules considered is FAR 36, from which we are going to apply requirements for a Stage 2 certification [18, 19]. Noise was not prioritised, since operations were unmanned and devised for a sea environment, far from human activities. However, reducing acoustic issues will prove helpful in a business case, in case of reconversion for other activities. Given the complexity of the models involved, it was not possible to perform an accurate computational noise analysis, due to time constraints and to the still incomplete vehicle configuration until the very last week of the project. It was deemed more sensible to have a qualitative analysis, supported by approximate estimations. However, in future iterations, numerical simulations and, most importantly, flight testing will be necessary to accurately determine our compliance. Admittedly, this section is meant as a form of outline of possible future solutions to the problem rather than a summary of results from the current design phase.

There are several noise sources in helicopter flight, which were summarised by Lowson [20] as follows:

- Engine noise: due to high frequency vibrations, usually it's not a primary concern and can be prevented via careful engine and structural design;
- High speed impulsive noise: given by shock formation at the tip of the blades during forward flight, it's the perfect example of the dependence of noise on the flight regime and on how a major design driver for helicopters is a low tip speed;
- Blade Vortex Interaction (BVI) noise: generated by the interaction of the blade with the vortex shed by the preceding one, which occurs mainly during manoeuvres or slow descents;
- Turbulence-induced noise: including ingestion of turbulence by the rotor, Blade-Wake Interaction (BWI) and direct radiation from the blades' boundary layer, with marked effects at the trailing edge;
- Main rotors discrete frequency noise: other sources of noise under more general flight conditions, e.g. blade thickness noise.

Tail rotor noise is not present in our synchrocopter. A careful design of the blades in future design iterations will inevitably involve some analysis to accurately depict noise effects due to geometry: a first important feature is given by the BERP geometry at the blade tip, known for the reduction of impulsive sound waves at high speed. Although still preliminary and subject to change, Yin [21] outlined in his report a first design for the tip shape, which should allow for more precise and hopefully promising results in the second iteration. Moreover, certain flight regimes will not be as influenced by this kind of noise, as the tip Mach number will be low enough to prevent transonic flow over the section. Our small rotor diameter is beneficial to have a lower tip speed at constant RPM, although this will be likely increased in future work and possibly lead to a rise in noise levels, if the RPM itself is not reduced. Another possible improvement would be to reiterate the aerofoil selection, if other sections were proven to be more effective in reducing thickness noise.

Particular concerns arise from our synchrocopter configuration when it comes to interactions of the

blades with not only vortices and wakes shed by the same rotor, but the mutual interactions of the intermeshing system has to be considered. Again, these cannot be modelled at this stage.

A very rough estimate of the helicopter Effective Perceived Noise level in dB (EPNdB) was provided by Lowson, referencing the work done by Perry and Pike, and given by Equation 2.

$$\text{EPNdB} = 91 + 10 \log_{10} \left[\left(\frac{V_T}{670} \right)^{7.8} \cdot \left(\frac{W}{31500} \right)^2 \times \left(\frac{B}{5} \right)^{-1} \times \left(\frac{A_B}{343} \right)^{-1} \times \left(\frac{V_C}{160} \right)^{3.3} \right] \quad (2)$$

This formula is extremely approximate and biased towards more conventional rotorcraft designs, but provides an initial value to work with in the next design iteration. The noise level was found to be 57.7 dB, which is much lower than the FAR requirement of 93.6 dB for our maximum takeoff weight. Please note that the MTOW was considered to be with the slung payload attached, consistently with the rest of the report, as a conservative measure. This value is most likely an underestimate, given the synchrocopter design, but it does provide an adequate margin for future work.

5 Conclusion

From a general perspective, the team was overall outstanding, responding very well to the coordinators' inputs and facing the design challenges with dedication and practical engineering judgement. Even the issues outlined in the first section were secondary, considering the difficulty of remote teamwork and the fact that a first iteration always involves mistakes and suboptimal design features. The next iteration should therefore include drastic improvements to the rotor design and refinements to performance calculations: the control and simulations teams would have more time to devise a functional control system and verify the flying capabilities of the vehicle, respectively. More accurate noise analysis should be incorporated into the vehicle design, paying particular attention to the BERP tip design feature and to the interaction of the blades with the complex set of vortices and wakes created by the intermeshing rotors.

Cooperation with the WRL team was often complicated but resulted in a functional overall design of the launcher and recovery vehicles, both able to comply with the mission requirements. The two different timelines the launcher and recovery teams were forced to follow created a few issues in terms of rushed and overengineered decisions and of communication, which should be addressed in future projects with separate teams cooperating with one another.

Regarding the internal management approach instead, the strategies adopted early in the project proved relatively successful, with a couple of exceptions: fluid and quick decision-making, flexibility and agility allowed the team to rationally face complex engineering challenges and to cooperate effectively and smoothly. This strategy was backed up by a low-level, centralised communication with only few intermediaries, subteam independence and trust towards individual members. Working groups are also significant as they allowed a compact, ordered development of macro-design features.

Finally, we calculated the base price of the helicopter as \$ 3,520,000 and the direct and indirect operating costs as 892 \$/FH and 300,640 \$/year, respectively. These are reasonably low and provide an excellent starting point for the business case, with the intention of minimising the mission costs in conjunction with the other team. The regression analysis and the neural network approach provided really close results, validating each other's predictions: these, however, should be refined by considering better optimisation metrics such as the relative error between real value and prediction, rather than the R^2 of the fit. Ideally, the regression dataset should be expanded to include as many helicopter models as possible, to further prevent overfitting and mainly to generalise for more unconventional designs. In conclusion, more precise and non-statistical methods of cost evaluation should be adopted, customised for the unconventional design we opted for. Given the complexity of a detailed approach, this should be done in a later design phase, perhaps by a dedicated subteam of two or three people.

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A Appendices

A.1 CAD model and general vehicle specifications



Figure 6: CAD model of SynchroCapture. Courtesy of Malhotra [?].

Table 3: Summary of vehicle specifications

Parameter	Value (Metric)	Value (Imperial)
Global Mass & Power		
Weight with Payload (MTOW)	2907 kg	6414 lb
Empty Weight, Helicopter	1229 kg	2709 lb
Fuel Capacity	278 kg	619 lb
Payload Mass	1400kg	3086 lb
Power Required (Ceiling, ISA)	1193.12 kW	1600 hp
Rotor (For one rotor)		
Number of Blades	2	2
Blade Mass (for each blade)	40 kg	88 lb
Rotor Diameter	8.05 m	26.4 ft
Rotor Aspect Ratio (AR)	14	14
Rotor Solidity	0.0909	0.0909
Rotor Disk Loading	45.9 kg/m ²	9.4 lb/ft ²
Rotor Blade Loading	503 kg/m ²	103 lb/ft ²
Rotor Hover Tip Speed	205.4 m/s	673.9 ft/s ($\approx$ 0.65 Mach at 20000 ft)
Rotor Cruise Tip Speed	268.529 m/s	881 ft/s
Rotational Speeds		
Engine Output Shaft	1256.6 rad/s	12000 RPM
Rotor Speed	51.1 rad/s	487.5 RPM
Travel Speeds		
Cruise Speed	185 km/h, 51 m/s	99 kts
Cruise Speed (Loaded)	100 km/h, 27 m/s	54 kts
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s	121 kts
Never Exceed Speed, v_{ne} (Tip Mach = 0.92)	243 km/h, 67 m/s	131 kts
Speed (Supersonic Flow, Tip Mach = 1)	264 km/h, 73 m/s	143 kts
Capture Velocity	144 km/h, 40 m/s	77 kts
Timings		
Total Mission Time	1 hr 49 mins	1 hr 49 mins

A.2 Team management - additional information

Parameter	Time for capture	Speed/Range	Cost	Likelihood of success	Pilot workload	Feasibility of design	Maneuverability and response	Normalized mean
Time for capture	1.00	3.00	0.20	0.14	8.00	0.14	1.00	0.44
Speed/Range	0.33	1.00	0.33	0.20	6.00	0.20	0.50	0.28
Cost	5.00	3.00	1.00	1.00	10.00	1.00	3.00	0.79
Likelihood of success	7.00	5.00	1.00	1.00	10.00	2.00	4.00	0.98
Pilot workload	0.13	0.17	0.10	0.10	1.00	0.10	0.13	0.06
Feasibility of design	7.00	5.00	1.00	0.50	10.00	1.00	6.00	1.00
Maneuverability and response	1.00	2.00	0.33	0.25	8.00	0.17	1.00	0.42

Figure 7: AHP matrix utilised for vehicle selection.

As also reported by Gliner [1] and Ghenima [2], the AHP matrix in Figure 7 was used to finalise the vehicle selection at the end of the first week of project: each design driver was compared to each other and was given a weight from 1 to 10 (from less important to more important). Values smaller than 1 logically are the inverse of these grades and indicate minor design objectives. The mean of each row was then normalised, giving a ranking of these drivers. As it can be seen, feasibility of design, likelihood of success and cost were deemed fundamental for a good overall design.

In Figure 8, a risk matrix was used to quantify the advantage of a vehicle compared to the others:

Parameter	Aircraft	VTOL	Helicopter
Time for capture	10.00	2.00	1.00
Speed/Range	1.00	5.00	10.00
Likelihood of success	6.00	4.00	2.00
Feasibility of design	3.00	6.00	4.00
Average	5.00	4.25	4.25

Figure 8: Risk matrix devised to further evaluate the convenience of each design during the vehicle selection.

some weights were given to each concept in order of risk with respect to each of the indicated design drivers. For example, the aircraft was considered highly risky in terms of time for capture, as it would have been difficult to have another attempt after a first missed recovery; instead, both helicopter and VTOL had an advantage due to the lower flight speed achievable. The lowest aggregate mean of these risk factors indicated the best design, in this case both VTOL and helicopter.

It should be noted that these are methodologies to quantify and justify a decision usually already made, or about to be, and not to make a choice irrespective of engineering judgement, since the weights given to the design drivers are somewhat arbitrary. In fact, the main reason for the choice of the helicopter, as mentioned in Section 2.3, was the precision in the recovery location given by the launcher team,

A.3 Cost estimation - additional information

A.3.1 Regression analysis

Here are three relevant plots of the direct operating costs against the maximum takeoff weight (Figure 9) and of the indirect operating costs against MTOW (Figure 10) and base price (Figure 11). These are useful to visualise the patterns relating these quantities. Other plots relating operating costs to range, altitude and speed have been omitted, since not as clear as these primary examples.

A.3.2 Neural network study

As mentioned earlier in Section 3.6, in Figures 12 and 13 the R^2 values and IOC predictions for two-layer networks as function of the number of neurons are presented. Both plots show similar features to those for DOC.

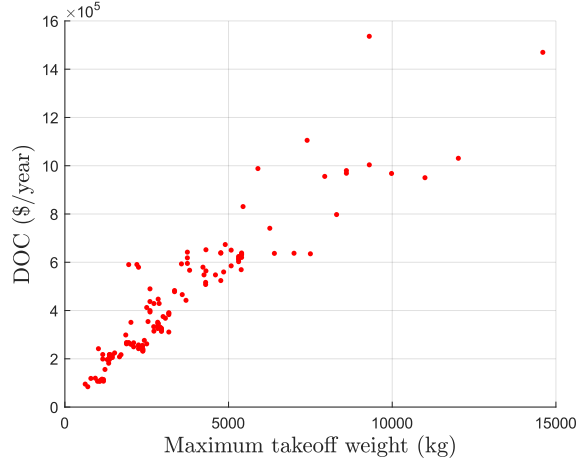


Figure 9: Direct operating costs against maximum takeoff weight.

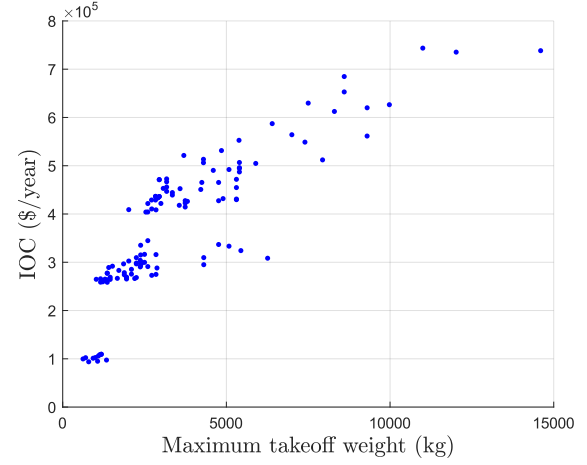


Figure 10: Indirect operating costs against maximum takeoff weight.

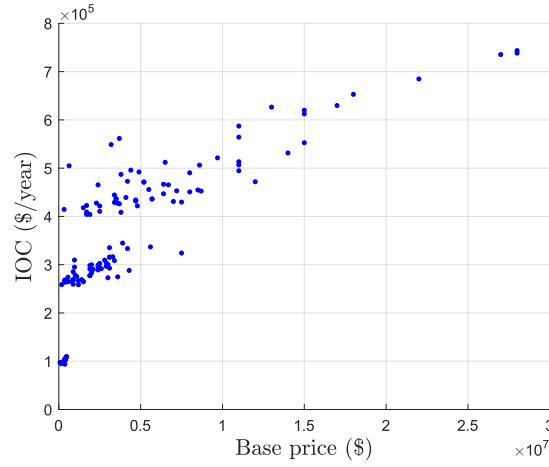


Figure 11: Indirect operating costs against base price.

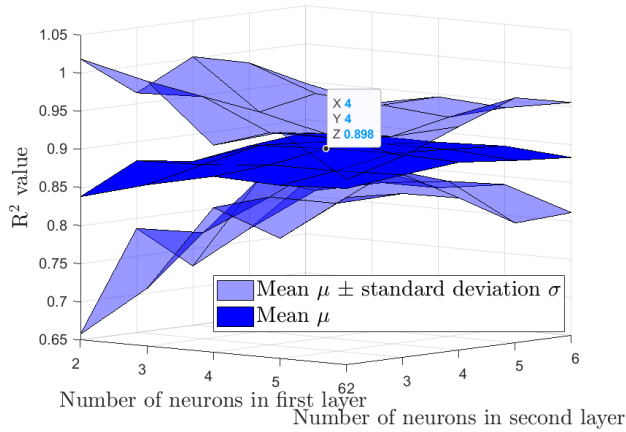


Figure 12: R^2 values of IOC predictions for two-layer networks as function of the number of neurons.

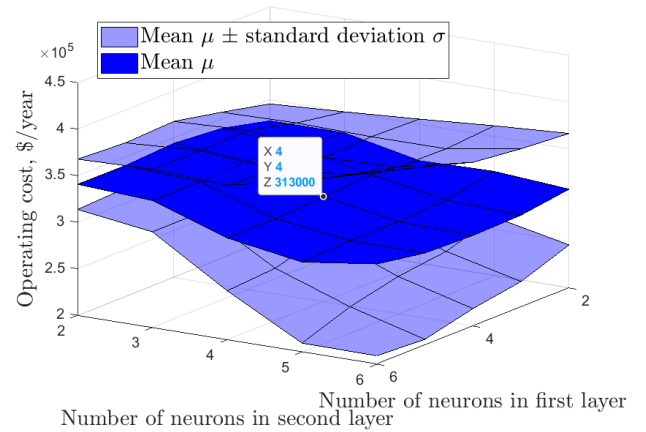


Figure 13: IOC predictions for two-layer networks as function of the number of neurons.